Orange City Council

West End Precinct
Development Control Plan

April 2021

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Definitions

Active frontages – a frontage is an active frontage if all premises on the ground floor of the building facing the street are used for the purposes of business premises or retail premises.

Amenity – is the 'liveability' or quality of a place which makes it pleasant to be in for individuals and the community. Amenity is important in both the public and private domain and includes the enjoyment of sunlight, views, privacy and quiet.

Apartment Design Guide - provides consistent planning and design standards for apartments across the State. It provides design criteria and general guidance about how development proposals can achieve the nine design quality principles identified in *SEPP 65 (State Environmental Planning Policy No 65 - Design Quality of Residential Apartment Development).*

Articulation– refers to the three dimensional design of a building and its surfaces. Building articulation can enrich the building's street address and character, and should respond to its orientation. Building articulation also includes modelling of the upper level and roof level of a building.

Building envelope – is the area within which a building can be built, usually represented in plan, section or 3d model.

Fenestration – refers to the design and placement openings in a building's envelope, most notably the windows, doors, skylights and curtain walls. This is mainly referred to as aesthetic but also effects the performance of the building.

Public domain - are places within an area which are available for public use and access, including streets, public spaces and open space.

Glossary

Abbreviation	Description
ADG	Apartment Design Guide – NSW Department of Planning, Industry and Environment
AJ&CA Design Report 2020	Allen Jack & Cottier Architects – Design Report (September 2020)
Council	Orange City Council
DCP	Development control plan
DPIE	Department of Planning, Industry and Environment
FOBH	Former Orange Base Hospital
FSR	Floor space ratio
GFA	Gross floor area
OLEP 2011	Orange Local Environmental Plan 2011
SEPP	State Environmental Planning Policy

1. Introduction

1.1 Land to which this Development Control Plan applies

This Development Control Plan (DCP) outlines site specific controls for the area identified as the West End Precinct located north of the Orange Central Business District (CBD) and adjacent to the western boundary of the new Department of Planning, Industry and Environment (DPIE) office and carpark complex.(Refer to Figure 1-1).

The DCP applies to the western undeveloped portion of the former Orange Base Hospital (FOBH). The property is described as Lot 501 in DP 1249083, 103 Prince Street, Orange. The site has a total area of 10,690 m².

The DCP will provide direction and manage the future medium density housing development proposed for the subject site as outlined in the FOBH Residential Development Design Report prepared by Allen Jack & Cottier Architects (AJ&CA) (Issue 5, September 2020).

The following zones apply to the land within the precinct as specified by the *Orange Local Environmental Plan* (OLEP) *2011*:

- R3 Medium Density Residential
- RE1 Public Recreation

The site is split into 3 separate areas providing an opportunity for multi-dwelling development integrated with public open space as outlined in Table 1-1.

Table 1-1 Land areas of R3 and RE1 zones

Zone	Street frontage	Area (m²)
R3 Medium Density Residential	Dalton Street (northern portion)	3,932
R3 Medium Density Residential	Prince Street (southern portion)	4,140
RE1 Public	Central portion of site	2,620

1.2 Relationship to other planning instruments and controls

Where any provision of this DCP is inconsistent with any provision of any other section of the Orange DCP, the provisions of this DCP shall prevail to the extent of any inconsistency. If any development matter is not addressed in this DCP, the controls in the Orange DCP should take effect.

1.3 Additional controls

A merit based approach to residential development in Orange allows developers and designers to respond to the individual opportunities and constraints of the site. This enables designers to respond to the character of the site with a flexible approach and maximise development potential and impacts on the surrounding neighbourhood.

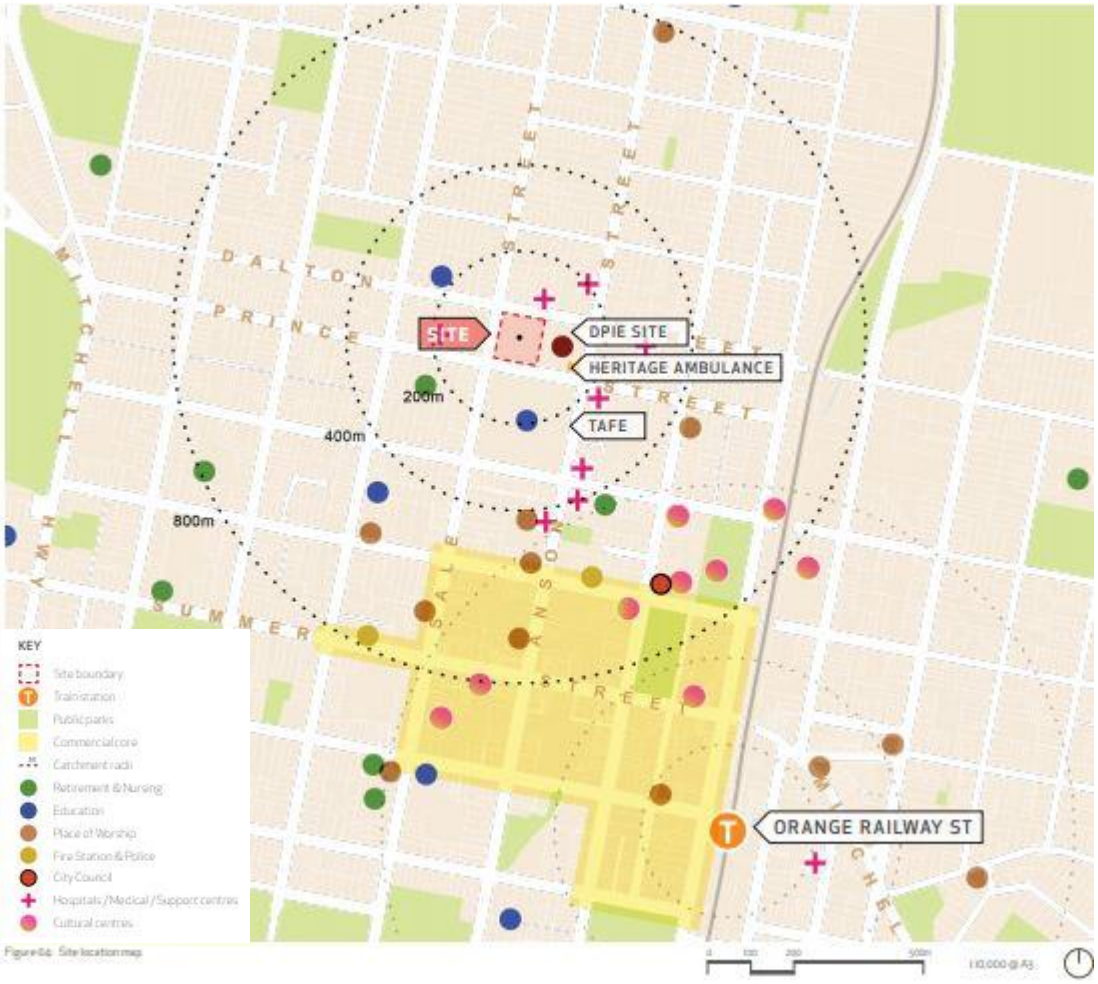


Figure 1-1 Land to which the DCP applies

The DCP promotes a comprehensive approach to the site's planning and design for new residential development. This requires analysis of the site's character including opportunities and constraints while considering the following design elements:

- Streetscape
- Energy Efficiency
- Bulk and Scale
- Privacy and Security
- Site Access and Circulation
- Water and Soil Management
- Open Space and Landscaping
- Site Facilities

The development controls for the precinct are not an exhaustive list of controls applicable to medium density housing and should be read in conjunction with other controls including those listed in the Orange DCP including:

- Part 03 – General Considerations
- Part 04 – Special Environmental Considerations
- Part 05 – General Considerations for Zones and Development
- Part 07 – Development in Residential Areas
- Part 13 - Heritage
- Part 14 - Advertising
- Part 15 – Car parking
- Part 16 – Appendices – Criteria Schedules & Conditions & Entrance and Plan maps

The following plans and strategies have guided the preparation of the controls:

- Orange Local Environmental Plan 2011
- Orange Development Control Plan
- OC Future City – Planning and Design Framework 2020

Reference should also be made to:

- SEPP 65 Design Quality of Residential Apartment Development
- SEPP Housing for Seniors or People with a Disability
- SEPP Affordable Rental Housing 2009

2. DA requirements

The following documentation is required to accompany a development application for the site.

2.1 Design details

New development planned for the site should include the following design details:

- Site plan: contours, boundaries, easements, existing trees, existing structures, drive and verge locations.
- Floor plan: plan for each floor, finished floor levels, schedule of gross floor area for each floor.
- Sections: finished floor levels from natural ground level, side and rear building envelopes.
- Elevations: dimensioned heights including overall height, finished floor level, ceiling height and finished ground level.
- Materials and finishes: palate including detail on the materials and finishes proposed as part of the development.

2.2 Car parking layout

A car parking plan is required for all multi dwelling, commercial and mixed use development and should indicate:

- Existing and proposed parking spaces off and on the site.
- Demonstrate how the parking generated by the development on the site meets the requirements of Council's *Orange DCP 2004 – DCP 15 Car parking.*

Note: the parking plan may be provided as part of the traffic report, if a traffic report is being submitted with the application.

2.3 Erosion and sediment controls plan

The following must be included as a minimum on the plan:

- Sediment controls (e.g. location of sediment fence).
- Stabilised access point.
- Cut/wash area.
- Water flow path (e.g. contours).

2.4 Landscaping plan

A landscape plan to be prepared by a qualified landscape architect and indicating:

- Proposed planting details, including plant species and common name, total number of each species of plant proposed and planting size.
- Trees on the site proposed to be retained and removed.
- Proposed landscaping.
- Tree protection zones for all protected trees affecting the site.
- Existing and proposed contours.

2.5 Perspectives

Perspectives are required to be supplied for multi dwelling, commercial, mixed use developments. Perspectives should show a 3D representation of the development in colour.

2.6 Pre DA community consultation report

Undertake consultation with the community prior to lodging a development application and prepare a report based on the feedback received.

2.7 Shadow diagrams

Shadow diagrams are required for all buildings or structures over three storeys in height or when a development does not comply with relevant rules for building or solar envelopes. A shadow diagram should show:

- Existing contours.
- Natural and finished ground levels related to Australian Height Datum.
- The location of proposed buildings and structures.
- Shadows cast by proposed buildings, buildings to be retained and proposed and existing fences at 9am, noon and 3pm on 21 June (the winter solstice).
- The location of all buildings and private open space on adjoining or adjacent blocks that are affected by those shadows.
- Elevations of buildings on adjoining or adjacent blocks that contain windows to habitable rooms affected by shadows cast. Where possible the type of habitable room affected should be indicated (e.g. bedroom, dining room).

2.8 Solar access plan

A solar access plan is required for multi dwelling and apartment developments to demonstrate compliance with SEPP 65 – Design Quality of Residential Apartment Development and the SEPP 65 Apartment Design guide. Solar access plans should also be provided for any townhouses.

2.9 Turning templates

Turning templates are required for developments of multi units (including dual occupancies), commercial, mixed use and industrial developments and must show:

- Manoeuvring of B99 swept path in accordance with *AS2890.1 Off-Street Car Parking.*
- Manoeuvring of 12.5 m waste vehicles.

2.10 Traffic report

A traffic report is to be prepared by a qualified traffic planner/engineer indicating traffic movement from Dalton, Prince and Sale Streets to and from the site.

2.11 Water sensitive urban design strategy

Water sensitive design documentation is required outlining sensitive design measures. A report is to be prepared by a suitably qualified professional.

2.12 Waste Management

A Waste Management Plan is to be prepared by a suitable consultant and written correspondence is to be provided to Council that indicates the proponent has liaised with the service provider.

3. Existing urban character

The West End Precinct is located within the established northern area of the City of Orange with frontages to Dalton, Sale and Prince Streets. The lands surrounding the site and DPIE are zoned R1 General Residential supporting detached single dwellings and a range of other mixed uses.

The DPIE office building is a four storey structure supported by a 5 storey carpark and is positioned on the western boundary of the precinct. The Orange TAFE complex is located to the south with frontage to Prince Street.

The Orange CBD is within close proximity of the West End precinct and within 10 minutes walking distance, highlighting the significant value of its location and opportunity to support multi dwelling housing within a short radius of the city's central core.

The wide streets and amenity surrounding the precinct have the potential for enhanced street beautification that could incorporate established trees within the existing streetscape. Sale Street provides connectivity opportunities with the CBD as it is supported by walking and cycling pathways and public transport linkages and bus routes.

The precinct is surrounded by low rise residential typologies utilised as medical and professional consulting establishments, other low impact business uses and residential purposes.

Dalton Street supports a wide median strip where established trees are located and provides amenity to the existing neighbourhood to the north.

The former Orange Ambulance Station is located adjacent to the new DPI building and is a heritage listed item under the OLEP 2011. There are also a number of heritage listed items surrounding the precinct with a heritage conservation area located to the south known as the Central Orange Conservation Area.

The southern portion of the precinct is relatively flat with a consistent slope. The northern portion of the site is predominantly flat, but has a steep embankment running along the western and northern edges supported by an existing retaining wall in the north-west corner.

The central portion of the precinct is relatively clear from vegetation and is reasonably flat particularly to the west. The highest point of the precinct is located in the north western corner and the lowest in the south eastern corner with a fall of approximately 4.2 metres. Existing vegetation is situated along the western boundary adjacent to Prince Street.

The precinct's opportunities and constraints are indicated below.

Table 3-1 Precinct opportunities and constraints

Theme	Opportunities	Constraints
Location	Located within close proximity and walking distance to the Orange CBD	Transport linkages including walking/cycle paths and bus routes may need formalising to better connect residents to the CBD precinct.
Topography	The precinct is relatively flat with the highest point in the north west corner.	On site water management will require consideration particularly in the southern portion of precinct.
Urban structure	The precinct is located adjacent to the DPI building, TAFE and within an existing established residential area.	Appropriate design and scale will be necessary to transition and integrate new development with existing built form.
Servicing	The precinct is included in the 30 Year Infrastructure Strategy as “Old Hospital Residential” (Site 5).	A servicing strategy will be required to be undertaken to determine accurate water and sewer Equivalent Tenement (ET) loads and calculations.
Infrastructure and road network	There are three (3) existing roads of considerable width that surround the site including Dalton, Prince and Sale Streets. There is opportunity to activate these street frontages.	Consideration of traffic movements and access from each road will be required to ensure safety for new residents to the precinct.
Landscaping	A number of existing mature trees are located at the western end of the precinct that will contribute to the creation of a central open space precinct.	The precinct is identified as having Groundwater Vulnerability. Consideration will be required when developing the site

4. Desired future character

The precinct's attributes provide an opportunity to create a medium density residential development supported by commercial uses and integrated with an open space plan.

4.1 Design principals

The following design principles are fundamental to the future development of the West End site.

4.1.1 Contribute to a clear, vibrant and safe street network

- Contribute to a vibrant street network while ensuring safe pedestrian and cycle movements throughout the site.
- Establish a new north-south pedestrian and bike-friendly link connecting Dalton and Prince Streets.
- Ensure that carpark entrances are appropriately located in accordance with the street network and slope of the site.

4.1.2 Ensure high amenity and safety to new public open spaces

- Provide safe and accessible entries to the open space at the centre of the site and include key elements including active and passive recreational spaces, walking and cycling paths, seating opportunities in sun and shade, security and lighting for safety.
- Ensure high amenity and safety to the new public open spaces for new and existing residents.
- Provide on-grade pedestrian links between the public park and the DPIE site.
- Ensure any new built form will not significantly overshadow publicly accessible open space.

4.1.3 Ensure massing and design of buildings is appropriate and sensitive to its surroundings

- Provide a high standard of functional multi dwelling residential development offering various densities that sympathetically relates to existing adjoining development and surrounding residential areas.
- Encourage innovative design with particular emphasis on the integration of buildings, open space and landscaped areas for community recreational uses.
- Ensure that the built form relates to the surrounding scale by establishing transitions in height.
- Provide buildings that are diverse in scale and articulation.
- Configure built form to ensure building entries are able to be clearly identified from public streets.
- Deliver a built form that delivers high yield on site without compromising amenity.

4.1.4 Design new built form that is climate responsive and carbon efficient

- Ensure new built form is best orientated to minimise removal of existing vegetation on site.
- Design apartment and townhouse layouts to maximise cross-ventilation and reduce reliance and use of air-conditioning.
- Ensure windows and walls are both well-insulated to maximum heat retention in winter and well-shaded to minimise heat gain in summer.
- Choose materials, colours and fixtures that are low in net carbon footprint and require minimal maintenance.

4.2 Objectives

The key objectives for the future development of the site are detailed below.

- Ensure a comprehensive approach to the site design for the effective functioning of residential and commercial development and integration of passive and active public open space.
- Ensure that new residential development complements the desired future character of the precinct typified in the AJ&CA Design Report 2020 which responds to the orientation, street pattern, street width, existing development, surrounding heritage buildings, street block size, land use and protects important public view corridors.
- Provide a range of housing typologies, building heights, architectural styles to create medium density housing diversity and visual interest.
- Building facades are to incorporate a variety of materials and textures, and relate sympathetically to existing buildings in the vicinity.
- Enhance the sense of place by respecting the surrounding heritage values.
- Ensure opportunities for housing choice and affordability in order to meet the broader housing needs of the community.
- Provide the opportunity for ground floor commercial uses within a mixed use development.
- Ensure new development is designed to minimise negative impacts on the surrounding amenity in relation to privacy and solar access.
- Create a complimentary mix of uses and building types integrated with open space.
- Encourage architectural outcomes that contribute to the local character.
- Provide connectivity and access to the site while retaining important views and vistas.
- Develop east- and north-south linkages with open space and existing road networks.
- Stimulate activity and a sense of community in public spaces.

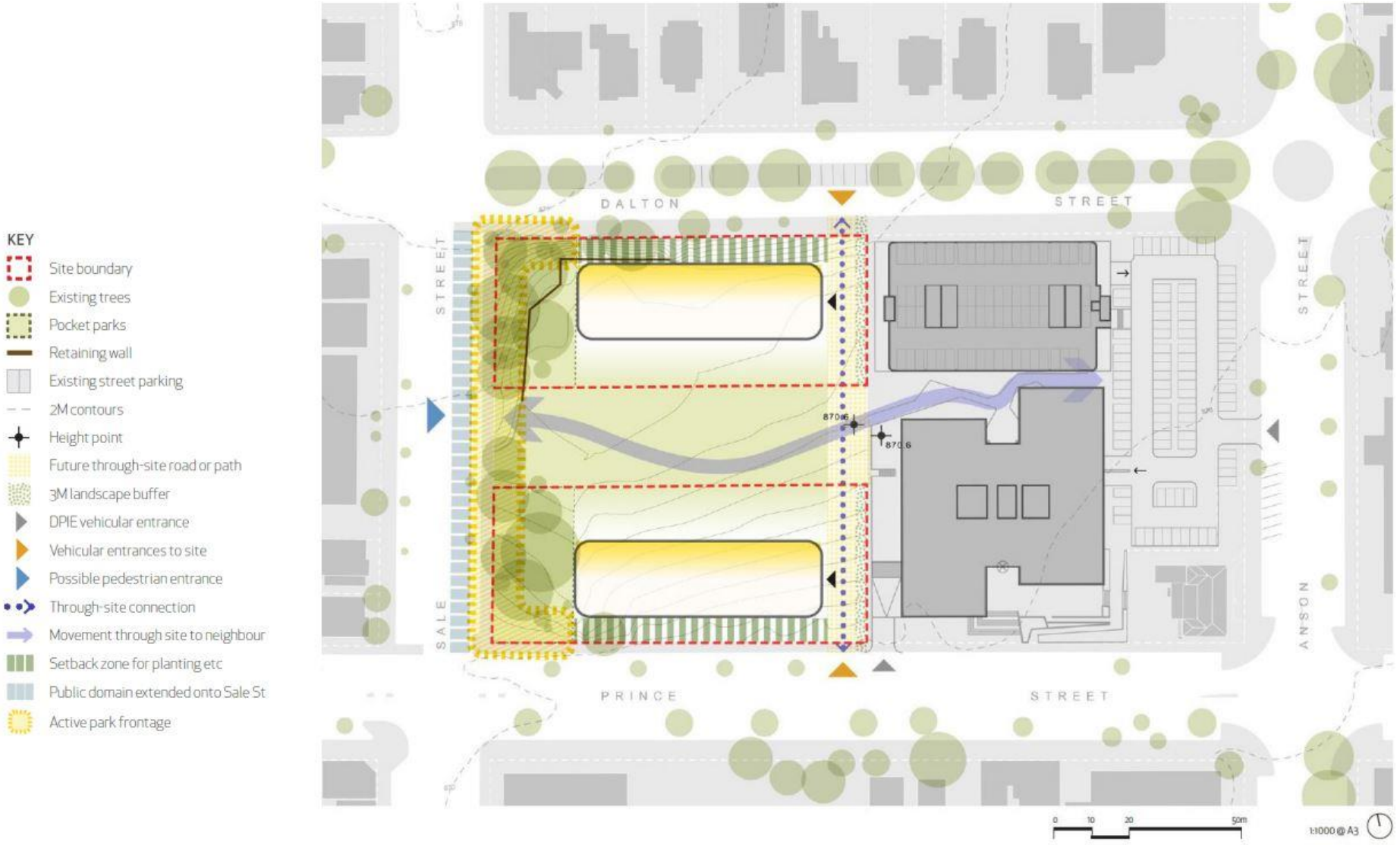


Figure 4-1 Design principles applied to the site

5. Infrastructure provision

5.1 Staged development

Proponents will be required to provide a 'concept proposal' (staging plan) for the development of the site. This should include detailed proposals for separate portions of the site that will be the subject of separate development applications.

The aim is to provide early acceptance of the development concept while providing stakeholder confidence. This will determine key design parameters including building envelopes, gross floor area (GFA), public and private space and access to ensure the issue of development consents are consistent with the initial concept design and approval.

5.2 Infrastructure, utilities and servicing

5.2.1 Objectives

- Ensure existing infrastructure is adequate to meet required levels of service, without impacting on existing amenity.

5.2.2 Controls

- Ensure compliance with Clause 7.11 of the OLEP 2011 including all requirements necessary for the provision of essential services including water, sewer management, stormwater drainage, electricity and telecommunications.
- Augmentation or enhancement of local infrastructure is to be at the developers cost and in accordance with the West End DCP.
- Location of local infrastructure is to be as per the requirements of the relevant agency or authority.
- A Servicing Strategy is required to demonstrate the availability and feasibility of providing water, sewer and stormwater services appropriate for the scale of development.
- Buildings and structures are to be located clear of utility infrastructure.
- Stormwater arrangements shall be designed to a gravity system.
- Refer to Council's Engineering Guidelines Developments for design requirements.
- The proponent is responsible for consulting with Essential Energy, natural gas and telecommunications agencies and authorities regarding the provision of services.
- Power lines are to be located underground where possible.

5.3 Excavation

5.3.1 Objectives

- Minimise the impact of excavation on the natural environment, neighbouring properties, and streetscape.
- Ensure minimal site disturbance and geotechnical stability of landfill and excavation works.

5.3.2 Controls

- New development is to respond to the local topography and slope of the site.
- Utilise the slope at the north eastern corner of site to locate servicing and parking, minimising its visual impact.
- Existing natural vegetation and trees should be retained and incorporated as landscape features on the site in order to maintain the natural character of the landscape.
- Trees are to be protected through any earthworks to prevent soil compaction and root damage.
- Fill is not to be used to raise the ground level.
- Avoid and minimise excavation where possible.
- If fill is to be transported on site and used the fill must be virgin excavated natural material (VENM).
- Fill must be managed on site, where possible, limiting fill moving off site.
- Ground levels for new development should reference the existing levels of the site, and any proposed levels that may arise from using fill or excavation.
- Earthworks undertaken within close proximity to the DPI building must protect the foundations.
- Consideration is required regarding appropriate excavation depths and potential impact on local groundwater vulnerability.

5.4 Public domain

5.4.1 Objectives

- Ensure the public domain area and assets are designed, managed and protected for future recreational community use.

5.4.2 Controls

- The public domain areas including streets, roadside reserves, and public open spaces are to receive a consistent level of embellishment to enhance the character of the precinct, in consultation with Council.
- Land identified for the RE1 Public Recreation Zone is to be dedicated to Council as public open space upon subdivision of the parent lot and embellishment.
- Outdoor furniture and open space embellishments area to be approved by Council.

Note: Refer to section 7.0 Public Domain including open space and parkland for further information

5.5 Security and lighting

5.5.1 Objectives

- Enhance public open spaces with lighting to ensure the safety of new residents, pedestrians and community members utilising the public domain area through the application of CPTED principles.

5.5.2 Controls

- Public open space area landscaping is to be configured to minimise concealment opportunities and maximise passive surveillance to encourage active street environments that enhance the sense of safety.
- Landscaping should consist of low ground cover with trees where low branches are avoided to provide a clear line of sight and avoids concealment opportunities.
- Public lighting is to be provided to all areas within the public domain to promote safety and provide an attractive environment at night. Such lighting is to be located and directed to minimise light spill into private areas.
- Enhanced levels of lighting are to be installed along the shared zone, within the open space area, pedestrian/cycle pathways.

6. Residential buildings

6.1 Built form, scale and articulation

6.1.1 Objectives

- Achieve diversity ensuring appropriate building scale and articulation.
- Ensure new development is sympathetic to adjacent built scale and density.
- Ensure buildings are designed for accessibility and aging in place.

6.1.2 Controls

General

- Consideration should be given to options for the inclusion of affordable housing and housing for seniors or people with a disability, within the precinct and building design.
- Configure built form to ensure building entries are accessible from the open space area and public streets.
- Figure 6-1 provides diagrammatic sections of the desired character including bulk and scale in context with the surrounding environment. New development will be required to reflect this concept.
- The number of dwellings that can be comfortably located on site are indicated in Table 6-1. Development applications will be required to design new development to reflect the number of apartments and townhouses indicated below.

Table 6-1 Indicative number of dwellings per hectare

Dwelling type	Apartments(south)	Townhouses (north)
Dwelling units	147 dwellings per hectare	49 dwellings per hectare

Apartments

- Position high amenity doubled-loaded apartments on the southern portion of the site to respond to the transition and adjacent built form including the DPIE site and TAFE site to the south.
- Provide sufficient communal open space along the west along Sale Street to preserve existing trees.
- Design apartments to address both Sale Street and Prince Street with access off both roads.
- Provide light and ventilation from multiple elevations to common lobbies and corridors.
- Ensure apartment orientation and layout is designed to maximise solar access and support cross ventilation.
- Provide dual aspect to apartment units where possible.

Townhouses

- Develop townhouses along Dalton Street to offer a finer grain of housing choice sympathetic in scale to existing low-density dwellings to the north (many of which are heritage listed).
- Open space supporting a public park to be located between both housing typologies with integrating features including passive and active recreation including playground and entertaining area.
- Ensure integration with the adjoining DPIE building and surrounding neighbourhood.
- North facing apartments should maximum solar access.
- Apply narrow building depths to maximise cross ventilation potential.



Figure 6-1 Section diagrams indicating built form, bulk and scale

6.2 Housing typology and density

6.2.1 Objectives

- The integrated design is to achieve a mix of medium density housing including town houses apartment types, sizes and orientations to increase housing choice.

6.2.2 Controls

- Achieve a combination of lower density townhouse development on the northern portion of the precinct facing Dalton Street with higher density double loaded apartment living to be located on the southern portion of the precinct facing Prince Street.
- New development must be capable of converting units to adaptable housing in accordance with *AS4299, Class C level.*
- The density and amenity for apartments and townhouse required to be achieved are indicted in Table 6-2.

Table 6-2 Housing typology, density and amenity

Housing Typology	Density	Amenity
Apartments	1 bedroom 2 bedroom 3 bedroom Top floor penthouse units	Large communal private open space Private open space Access to street
Townhouses	3 bedroom + study	Individual street address Central courtyard voids Private lock up garage Front and rear gardens

Apartments

- Design a mix of units to meet the housing needs of the community as indicated in Table 6-3. *Noting that the mix of dwelling below exceeds 100% to provide flexibility dependent on market conditions.*

Table 6-3 Mix of units to be included in apartment building

Dwelling Type	Dwelling Design
1 bedroom units	20%
2 bedroom units	70%
3 bedroom units	15%

- Provide a range of cross-through, dual aspect and corner units to provide options and choice for higher density apartment living.
- All apartments are to be designed to meet or above the requirements specified in the *Apartment Design Guide* (ADG)
- Ensure the living/dining are designed to the standards outlined in the ADG.

Townhouses

- Townhouses should be designed to the density indicated in table 6-1 Indicative number of dwellings per hectare.
- Living and dining areas, located on Level 1 (street level) should have a north-facing porch and entrance from the street with bedrooms positioned above.

- Private open space should be allocated as a front garden in keeping with the current character of the Dalton Street frontage.
- Generous private open space to be provided for the two end townhouses on the western edge fronting Sale Street.
- Each residence should be allocated at least one dedicated car space with convenient access to the main living quarters on ground and level 1.
- Garages and car parks are sited and designed so that they do not dominate the street frontage.

6.3 Building setbacks

6.3.1 Objectives

- Setbacks should complement the streetscape and provide sufficient area for landscaping to complement building form.
- Provide transitional spaces that enable engagement between public and private domains.
- Minimise any adverse impacts on adjoining properties such as privacy and overshadowing.

6.3.2 Controls

- Setbacks of proposed development are to comply with Table 6-4 Building setbacks.
- Townhouses facing the Dalton Street are to have individual entries from street level.

Table 6-4 Building setbacks

Street	Setback in metres (m)	Comment
Dalton Street	6.5	Setback aims to support the strong trees lined median strip along Dalton Street to the north and new landscape planting.
Prince Street	6.5	Setback will provide for landscaping on Prince Street to support new building form and any proposed ground floor commercial uses.
Sale Street	21.4	Create an open space buffer between proposed development and Sale Street.
DPI building and existing carpark.	3	Provide a buffer between existing business development and proposed residential development. A 10.5 metre width from this 3 metre buffer is required to be allocated to create a link through road through the site from Dalton Street to Prince Street.
Setback above 4 storeys	5	This setback is to be measure from the building line and ensures the bulk of the building is minimised.

6.4 Private open space and privacy

6.4.1 Objectives

- Ensure privacy is provided between buildings and avoid overlooking of living spaces and private open space.

6.4.2 Provisions

- Private open space must be provided for apartments and townhouses on ground level in relation to orientation for solar access.
- Private open space on ground level for apartments and townhouses must be fenced. Details of the height and style of fencing must accompany the development application.
- Should be directly accessible from a living area and include an alfresco area, balcony or similar at ground level.
- Multi-storey development must locate and size windows to habitable rooms to avoid facing onto windows, balconies or courtyards of adjoining dwellings.
- Windows of any bathroom, ensuite or WC should be treated with frosted or obscure glass.
- Direct views from living rooms of dwellings into open space or the interior of other dwellings are to be screened or obscured.
- If private open space is located on a corner, details of fencing should be provided.
- Balconies for private open space should meet the requirements of the ADG
- Private open space should be designed so as not to overlook into habitable rooms, outdoor entertainment spaces or courtyards.
- Air conditioning units are to be located so that noise measured at the property boundary does not exceed 5dB above ambient night-time background noise.

6.5 Solar access and overshadowing

6.5.1 Objectives

- Ensure that development does not unreasonably diminish sunlight to adjoining properties including open space areas.

6.5.2 Controls

- Ensure the configuration of buildings and structures must ensure that minimum of 50% of the area of public open space receive a minimum of 3 hours on 21 June (mid-winter).
- New development including the townhouses in the north and apartment building in the south should not reduce the solar access to neighbouring apartments of townhouses.
- Ensure compliance with the ADG in relation to overshadowing of neighbouring apartments.
- Position apartments and townhouses to the north for maximum solar access where possible.
- Maximise adequate solar access and natural lighting to internal and open spaces.
- Shadow diagrams are required for developments ≥ 2 storeys and need to demonstrate habitable rooms of adjoining dwellings and major part of their landscaped open space to retain a minimum of 4hrs sunlight between 9am-3pm on 21st June (winter solstice). *Note: "habitable" refers to rooms capable of occupation and does not include laundry, bathroom or garages.*

6.6 Building services

6.6.1 Objectives

- Provide and integrate site services and facilities in a sensitive manner so they integrate with the building and landscape design, enable easy access, and require minimal maintenance.
- Minimise visual impact by encouraging building services to be located on the ground floor/basement of buildings, where practicable.

6.6.2 Controls

- Ensure that building services are integrated into the design of buildings. Building service elements include garbage rooms, mailboxes, fire hydrant boosters, electrical substations, downpipes, and plant rooms and satellite/communications structures.
- Provide mailboxes adjacent to the main entrance and integrated into a wall of the building where possible, ensuring that they are secure and can accommodate large articles such as newspapers.
- Provide adequate space and facilities for outdoor communal clothes drying.
- Locate any ancillary structures such as plant rooms and satellite dishes away from the building entry, communal and private open spaces, and bedrooms.

6.7 Waste Management

6.7.1 Objectives

- To provide domestic waste management systems that allow for ease of use by occupants and safe and efficient service by collection contractors.

6.7.2 Controls

- A waste management plan is required for any application that details the number of waste, recycling and organics bins to be provided.
- Written evidence is to be provided to Council indicating that the proponent has conversed and co-designed servicing areas that are compatible with the servicing needs of the provider.
- Adequate storage for waste materials must be provided on site.
- Waste storage areas and facilities are to be convenient and accessible to the occupants of all dwellings.
- Waste storage areas should be incorporated into the design of the development and not be visible from the street.
- Waste is to be managed within the built form through the basement or suitably designed waste rooms so that there is no visual or odour impacts on the amenity of the shared zone or neighbouring sites.

7. Public domain

7.1 Open space

- Establish a vibrant community open space precinct in the centre of the multi dwelling development to encourage passive and active recreation.

7.1.1 Controls

- Public open spaces that are to be dedicated to Council are to be designed in consultation with Council and appropriate embellished at the developers cost.
- The following types of open spaces are required to be considered and included in the design for multi dwelling development and indicated in Figure 7-:
  - Layered open spaces-private, communal, public and street.
  - Private open spaces abut the public park to enhance the perceived depth of space available.
  - Public Playground for new and surrounding residents.
  - Barbeque and entertainment spaces.

Public Park

- The design and embellishment of the park is to be undertaken in consultation with Council that considers the following:
  - Provide a hard surfaced pathway that traverses the open space area and links into embellishments such as shelters, seats and barbeques areas.
  - Vegetation on-site should be retained where possible.
  - Suitable materials from excavation may be integrated into the design as seating and edge treatment.
  - Landscaping of the park shall be designed in consultation with Council to identify opportunities for local art and sculpture to contribute to the sense of place.
  - The implementation of shade structures for casual passive recreation will ensure a variety of activities in the park.
  - Provide on-grade pedestrian links between the public park and the DPIE site.
  - Provide a new pocket park to the western edge of the site to retain as many of the existing trees on site as possible.
  - Communal and private open spaces are to be designed to leverage the adjacent public park.
  - Include access provisions and facilities consistent with the requirement for seniors and people with a disability.
  - Locate bollards, barriers, perimeter trees and/or fencing to prevent unauthorised vehicular access into the open space.
  - Maintain clear sight lines across passive and active use areas.
  - Design open space area and facilities to maximise energy and water conservation.
  - Select materials and furniture items that are easily maintained, cleaned and replaced if damaged, as per Council requirements.

Playground Area

- An area within the open space precinct is to be provided for an active and passive recreation, including nature play and enhanced with locally sourced products and materials, as per Council requirements.
- Sculptural objects and seating can be interspersed in and around the playground to promote passive activities and provide seating for parental supervision.
- Where appropriate and safe, any materials found on site could be re-purposed as playground features which could include excavated stone as platforms to equipment or trees as seating or postmarks.
-



Figure 7-1 Playground Precedents

Top Left: Caloundra Cristian College Junior School Play Space, by Green edge Design Consultants

Bottom Left: Ollinda Playspace, by Fitgerald Frisby Landscape Architecture

Right: Railway Park, by Plummer & Smith

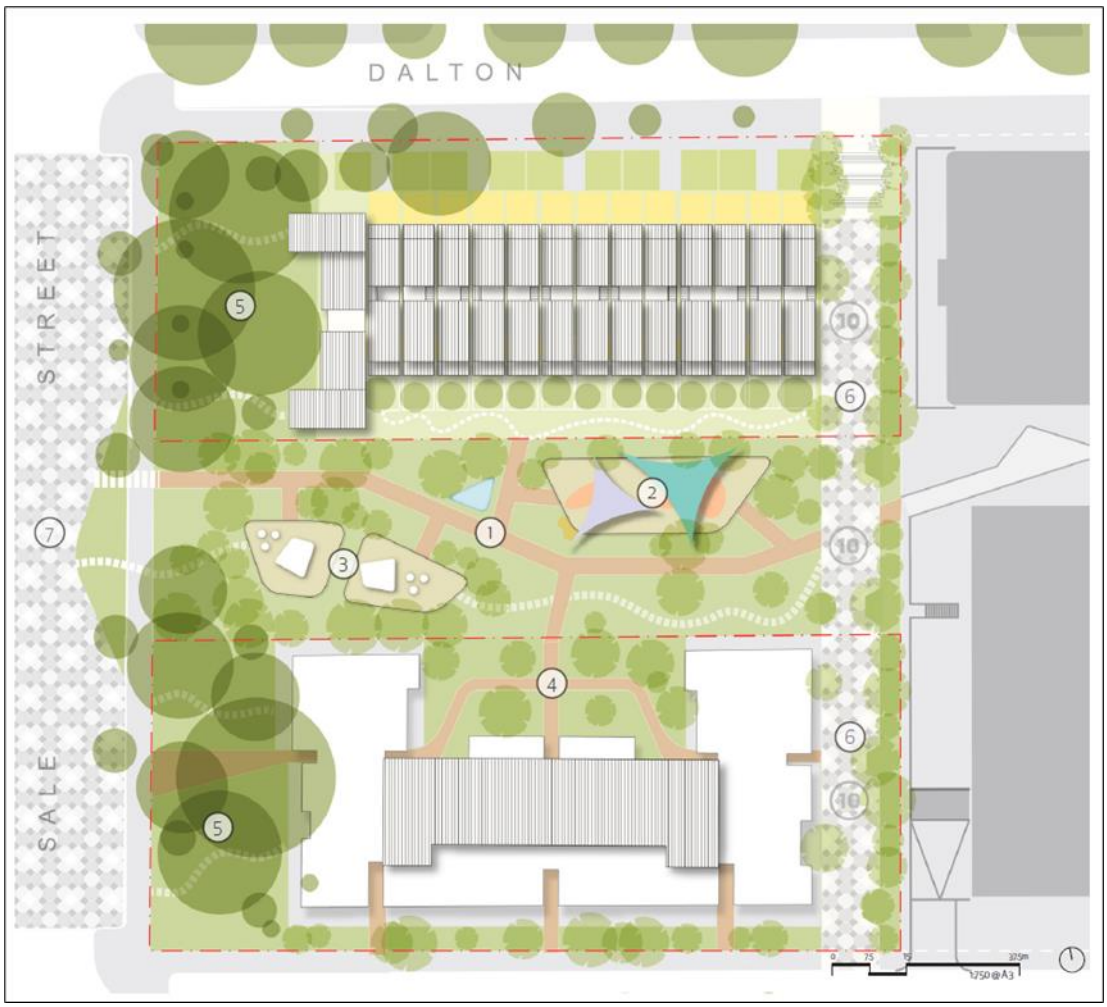


Figure 7-2 Indicative open space plan

The spaces include:

1. Public Park
2. Playground
3. BBQ Space
4. Residential communal open space
5. Residential Private space
6. New shared zone
7. Sale Street

7.2 Landscaping and character

7.2.1 Objectives

- Provide a landscape design that complements the nature and scale of the development and contributes to the desired streetscape character.
- Improve amenity and soften the impact of buildings through a landscaped environment including the retention and/or planting of vegetation.
- Assist with site hydrological management and water quality.

7.2.2 Public Domain

- Retain the existing deep soil zones to support established vegetation including mature trees and retain hydrological structure of the site.
- The calculation of deep soil zones may be co-located with existing vegetation and/or new plantings and include permeable paving or decking.
- Significant trees along the western boundary of the site are to be maintained to create a new north-south park at the end of RE1 Public Recreation zone.
- Street trees are to be provided along Dalton and Prince to provide a landscape buffer, with appropriate landscaping, as per Council requirements.
- Landscape design is to be integrated with water and stormwater management.
- The open space precinct is to provide for passive recreation for the surrounding residents and workers in the area, including but not limited to:
  - Tree planting for shade where required.
  - Low planting.
  - Hardscape areas for gathering.
  - An informal natural play area for children.
  - Seating.
  - Water Sensitive Urban Design to minimise use of potable water.
- Green walls are to be provided, where suitable to offset the urban heat island effect as well as limit the perceived bulk of the built form.

8. Movement networks

8.1 Street networks and site connectivity

8.1.1 Objectives

- Facilitate convenient movement through the site via street networks while ensuring safe pedestrian and cycle connections.
- Create strong pedestrian linkages between Sale Street and Anson Street and shared zone between Dalton Street and Prince Street.
- Provide active frontages, where appropriate, along Prince and Sale Streets and connectivity to public open space to encourage a vibrant urban environment that facilitates community activity and safety.
- Provide articulated frontages along Dalton, Prince and Sale Streets.

8.1.2 Controls

- Establish a new north-south pedestrian and bike-friendly lane connecting Dalton and Prince Streets.
- Footpaths along Dalton, Prince and Sale Streets are to encourage walking and cycling activities.
- Provide safe and accessible entries to the proposed recreational park at the centre of the site that will support passive recreational areas, event spaces, walking and cycling paths, a variety of seating opportunities and playground area.
- Create new pedestrian link from RE1 Public Recreation zone to neighbouring DPI site.
- Accessible, visible and friendly approach to large park at RE1 zone from Sale, Dalton and Prince streets, as well as via new link from neighbouring site.

8.2 Site access, shared zones and driveways

8.2.1 Objectives

- Ensure that vehicular access to and from the development is simple, safe and direct.
- Minimise adverse on-site traffic impacts and create pedestrian safe environments.

8.2.2 Controls

Pedestrian access

- Public pedestrian access to the public park is gained from the following as indicted in Figure 8-1:
  - Sale Street.
  - Through site connection at grade from the DPI site.
  - Prince Street via the shared zone laneway.
  - Dalton Street via the shared zone laneway.

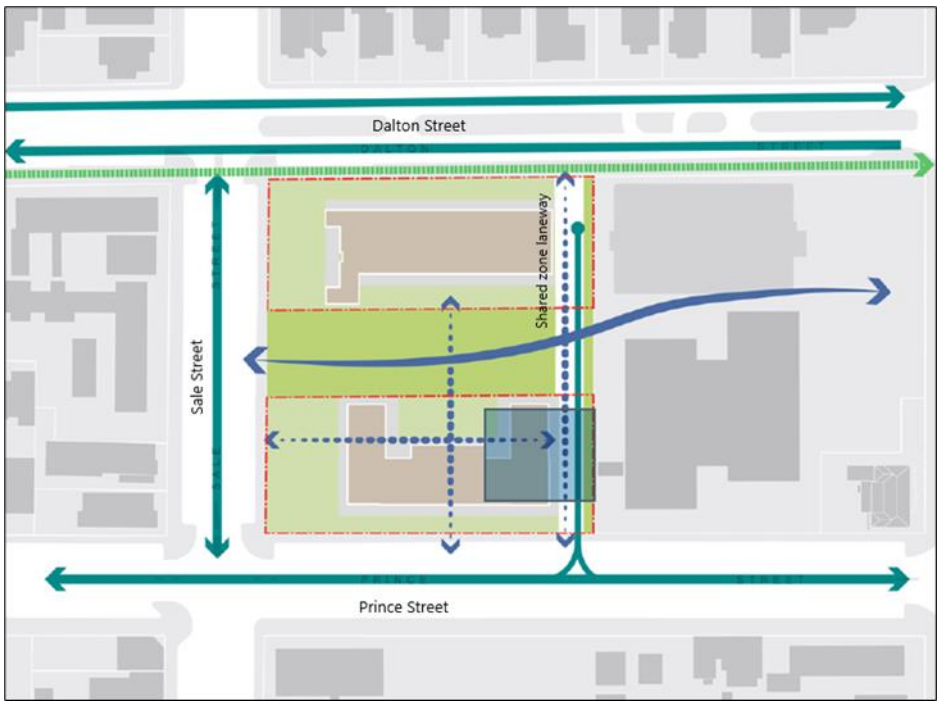


Figure 8-1 Pedestrian and vehicle movements

Vehicular access

- Vehicular access is gained from Prince Street.
- Driveway design shall provide safe and efficient ingress and egress to the site.
- The design of driveways and parking areas shall minimise the visual impact of hard paved areas.
- The driveway design shall make provision for service vehicles where practicable.
- All driveways and open car parking spaces and access ways should be suitably landscaped and lit to enhance the amenity of the site and provide security.
- Car parking areas and access ways should be designed, surfaced and sloped to facilitate stormwater infiltration on the site.
- Asphalt is required on approaches to the share way zone, consistent with existing treatment.

Parking access

- Basement parking for the apartments in the southern section of the site are to be accessed from the new shared zone laneway off Prince Street.
- Townhouse parking in the northern section of the site is to be accessed via the new shared zone laneway off Prince Street.
- The apartment lobbies may be accessed by pedestrians from multiple points at Prince Street, Sale Street, the new shared zone laneway and the communal open space.
- Townhouses are to be provided with individual access off Dalton Street and may include secondary pedestrian access to the public open space area.
- The western townhouses may be accessed off Sale Street, provided the established trees are retained.

- Vehicular access is to be designed to give priority to pedestrians and bicycles by continuing a similar type of footpath material and grade.
- Traffic calming treatments are to be incorporated into the onsite shared zone roadway to slow vehicles movements and ensure sufficient safe crossing points.

8.3 Visual privacy and safety

8.3.1 Objectives

- Ensure residential amenity is provided within and between new residential developments and open space area.
- Minimise opportunities for criminal and anti-social behaviour through the implementation of CPTED principles.

8.3.2 Controls

- Development should be designed to provide clear sightlines and lighting between public and private places.
- Ground level open spaces must not directly look into rooms and private landscaped areas of adjoining properties unless screening can mitigate privacy.
- Dwellings should be oriented towards the street with entrances and street numbering clearly visible.
- Multi-storey development must locate and size windows to habitable rooms to avoid facing onto windows, balconies/courtyards of adjoining dwellings.
- Bollard lighting is to be incorporated into the entrance of walkways and spaced along walkways to reduce antisocial use and remove potential concealment opportunities.
- Clear lighting around the approaches to doorways and windows enhance monitoring.
- Landscaping should consist of low ground cover with trees where low branches are avoided to provide a clear line of sight and avoids concealment opportunities.
- Fence heights are to be no higher than 1.2 m.
- Public entrances to buildings are to be distinct and of a scale consistent with the overall building.
- On-site parking is to be provided as per the *Orange DCP 2004 – DCP 15 Car parking*.
- Commercial premises located in the apartment building should have an active frontage to Prince Street to manage patron security.
- New development must ensure CPTED principles have been applied and informed the design (refer to the NSW Crime Prevention through Environmental Design Guideline).

9. Car parking

9.1 Parking controls

9.1.1 Objectives

- Provide adequate and convenient onsite parking for residents and visitors vehicles to maintain the amenity of the adjoining properties and the efficiency of the road network.
- Provide for adequate parking for commercial uses that may locate on the site.

9.1.2 Controls

Residential development

- Car parking is to be provided as per *Orange DCP 2004 – DCP 15 Car parking.*
- Ensure that carpark entrances are appropriately located in accordance with the street network and slope of the site.
- Car entry to the apartment building basement is to be accessed from inside the subject site.
- Basement parking is to be provided to apartments, with townhouses to accommodate under-croft parking utilising the existing slope to minimise excavation.
- Basement level parking is to be provided along with access to the apartments above and sufficient area for pedestrian circulation.
- The apartment basement should not exceed the structural footprint so as not to encroach into the deep soil zone to allow for adequate space for landscaping and tree planting.
- Townhouses to be designed to provide under-croft parking using the site slope to minimise excavation.
- Where the topography of the land or constraints of the groundwater table result in the basement parking level projecting above ground level, it is to be designed to:
  - Not project more than 1.2 m above ground.
  - Achieve an attractive ground level relationship between the building(s) and the public domain.
- Garages and parking structures are not to project forward of the building line.
- Design of car parking areas must ensure priority is given to the needs or pedestrians, disabled people and cyclists above the needs of the car.

Commercial development

- Separate car parking is required to service the commercial development located on the ground floor of the apartment building facing Prince Street. Car parking rates are to be provided as quoted in the *Orange DCP 2004 – DCP 15 Car parking.*
- Loading and unloading areas for retail and commercial development are to be designed and screened from public roads.



Figure 9-1 Indicative car parking layout

10. Environmental management

10.1 Environmental heritage

10.1.1 Objectives

- Ensure new development within the vicinity of heritage listed items and in proximity to the Heritage Conservation Area reflects and promotes an understanding and appreciation of the heritage significance and setting.
- Protect attributes which contribute to the heritage value and character of the area.

10.1.2 Controls

- Ensure consistency with the objectives of the Orange DCP 13 – 'Heritage' in relation to heritage conservation and management of new residential and commercial development.
- Ensure new buildings do not compete in height and scale with existing historical character including the former Orange Abundance Station and surrounding dwellings. Consideration must be given to the relationship of the development adjoining the DPIE building.
- Ensure new development respects the required building setback to maintain existing historical streetscape character.
- Interpretive design to new apartment structures is to ensure it is respectful to the historical character of Heritage Conservation Area located on the northern fringe of Prince Street.
- Built form design can mirror or interpret surrounding heritage items and pay homage to the heritage elements through building design, i.e. fenestration and materiality.
- Materials for new buildings should be in context with the surrounding brick work, masonry and roofing products.
- Maintain space between the building line and front fence and ensure garages and carports are located away from streetscape to maintain character of residential areas.
- Maintain and enhance public views through streetscapes and to the heritage conservation area to the south of Prince Street.

10.2 Water and energy efficiency

10.2.1 Objectives

- Protect and maintain the hydrological functions of the key groundwater systems.
- Apply Integrated Water Cycle Management (IWCM) principles for the provision of water services to support IWCM targets.
- Apply Water Sensitive Urban Design principles to mitigate water security.
- Protect the environment by maintaining and/or improving the quality of stormwater run-off.
- Apply sustainable energy efficient design to minimise non-renewable energy usage of new buildings.

10.2.2 Controls

Groundwater Vulnerability

- The site is identified as having ground water vulnerability as indicated by the OLEP 2011 Groundwater Vulnerability Map.
- New development should avoid or mitigate any significant adverse environmental impact on the hydrological functions of the groundwater system.
- Avoid or mitigate potential polluting on site activities to minimise risk to groundwater contamination.

Integrated Water Cycle Management

- Ensure the effective and efficient delivery of water and sewer services to the site while promote sustainable water conservation practices for new residents and managing water demand.
- Manage urban stormwater on site and encourage harvesting for reuse within the open space precinct and landscaping areas.

Water sensitive urban design (WSUD)

- Integrate WSUD with surrounding street network and open space area to facilitate stormwater management to manage volumes, quality and runoff velocities to pre-development levels.
- On-site storm water detention will be required across the whole site.
- Ensure water moving off-site is of an appropriate quality.
- Off-site water is to be of good quality, any water moving offsite requires a buffer ensure appropriate water quality.
- Raingardens should be incorporated into public open spaces to manage runoff speeds and water quality.
- Irrigation of the open space area should be undertaken using harvested stormwater captured on site.
- A WSUD strategy is required to be prepared for the site and include best practice WSUD elements.

Energy efficiency

- Apply the principles and properties of thermal mass, glazing, insulation and solar energy are to be recognised and incorporated into the design of residential development.
- Reduce reliance on artificial lighting and design lighting systems to target only those spaces which require lighting at any particular ‘off-peak’ time, not the whole building. Incorporate a timing system to automatically control the use of lighting throughout the building.